

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{1}{3} (\Upsilon_1 k^2 + 3 \overset{(2)}{\kappa_2}) & \frac{1}{2} (\Upsilon_2 k^2 + 2 \overset{(2)}{\kappa_2}) \\ \hline \end{array} \quad \mathcal{K}_{0^+}^{\#2} \begin{array}{|c|} \hline \frac{1}{2} (\Upsilon_2 k^2 + 2 \overset{(2)}{\kappa_2}) \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline \frac{1}{2} (\Upsilon_2 k^2 + 2 \overset{(2)}{\kappa_2}) & \Upsilon_3 k^2 + \overset{(2)}{\kappa_2} \\ \hline \end{array} \quad \mathcal{K}_{1^-}^{\#1}{}_\alpha \quad \mathcal{K}_{1^-}^{\#2}{}_\alpha \\
\mathcal{K}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{1}{3} (\Upsilon_4 k^2 + \overset{(2)}{\kappa_2}) & \frac{1}{6} (\sqrt{5} \Upsilon_5 k^2 + 2 \sqrt{5} \overset{(2)}{\kappa_2}) \\ \hline \end{array} \\
\mathcal{K}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{1}{6} (\sqrt{5} \Upsilon_5 k^2 + 2 \sqrt{5} \overset{(2)}{\kappa_2}) & \frac{1}{3} (\Upsilon_6 k^2 + 5 \overset{(2)}{\kappa_2}) \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{K}_{3^-}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \mathcal{K}_{3^-}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -\frac{k^2 \overset{(4)}{\kappa_3}}{3} \\ \hline \end{array}
\end{array}$$

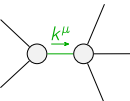
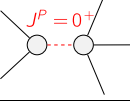
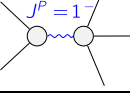
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{\Upsilon_3 k^2 + \overset{(2)}{\kappa_2}}{\text{Det}(0^+)} & \frac{-\Upsilon_2 k^2 - 2 \overset{(2)}{\kappa_2}}{2 \text{Det}(0^+)} \\ \hline \end{array} \quad \mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline \frac{-\Upsilon_2 k^2 - 2 \overset{(2)}{\kappa_2}}{2 \text{Det}(0^+)} & \frac{\Upsilon_1 k^2 + 3 \overset{(2)}{\kappa_2}}{3 \text{Det}(0^+)} \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \quad \mathcal{J}_{1^-}^{\#1}{}_\alpha \quad \mathcal{J}_{1^-}^{\#2}{}_\alpha \\
\mathcal{J}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{\Upsilon_6 k^2 + 5 \overset{(2)}{\kappa_2}}{3 \text{Det}(1^-)} & \frac{-\sqrt{5} \Upsilon_5 k^2 - 2 \sqrt{5} \overset{(2)}{\kappa_2}}{6 \text{Det}(1^-)} \\ \hline \end{array} \\
\mathcal{J}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{-\sqrt{5} \Upsilon_5 k^2 - 2 \sqrt{5} \overset{(2)}{\kappa_2}}{6 \text{Det}(1^-)} & \frac{\Upsilon_4 k^2 + \overset{(2)}{\kappa_2}}{3 \text{Det}(1^-)} \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{J}_{3^-}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \mathcal{J}_{3^-}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -\frac{3}{k^2 \overset{(4)}{\kappa_3}} \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
\Upsilon_1 &= 3 \overset{(4)}{\kappa_1} + 3 \overset{(4)}{\kappa_2} + 2 \overset{(4)}{\kappa_3} + 3 \overset{(4)}{\kappa_4} \quad \&\& \Upsilon_2 = 2 (\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}) + \overset{(4)}{\kappa_4} \quad \&\& \Upsilon_3 = \overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2} \quad \&\& \Upsilon_4 = \overset{(4)}{\kappa_2} + \overset{(4)}{\kappa_3} + \overset{(4)}{\kappa_4} \quad \&\& \\
\Upsilon_5 &= 2 \overset{(4)}{\kappa_2} + \overset{(4)}{\kappa_4} \quad \&\& \Upsilon_6 = 5 \overset{(4)}{\kappa_2} - \overset{(4)}{\kappa_3} \quad \&\& \text{Det}(0^+) = \frac{1}{12} (k^4 (8 (\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}) \overset{(4)}{\kappa_3} - 3 \overset{(4)}{\kappa_4}^2) + 8 k^2 \overset{(4)}{\kappa_3} \overset{(2)}{\kappa_2}) \quad \&\& \\
\text{Det}(1^-) &= \frac{1}{36} (k^4 (16 \overset{(4)}{\kappa_2} \overset{(4)}{\kappa_3} - 4 \overset{(4)}{\kappa_3}^2 - 4 \overset{(4)}{\kappa_3} \overset{(4)}{\kappa_4} - 5 \overset{(4)}{\kappa_4}^2) + 16 k^2 \overset{(4)}{\kappa_3} \overset{(2)}{\kappa_2})
\end{aligned}$$

Lagrangian

$$\begin{aligned}
&\overset{(2)}{\kappa_2} \mathcal{K}_\alpha^{\alpha\beta} \mathcal{K}_{\beta\chi}^{\chi} + \overset{(4)}{\kappa_1} \partial_\beta \mathcal{K}_\chi^{\delta} \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} + \overset{(4)}{\kappa_2} \partial_\chi \mathcal{K}_\beta^{\delta} \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} + \\
&\overset{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^{\delta} + \overset{(4)}{\kappa_4} \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} \partial_\delta \mathcal{K}_{\beta\chi}^{\delta} - \frac{1}{3} \overset{(4)}{\kappa_3} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} = 0$	5	$k^\chi (\partial_\epsilon \partial_\chi \partial^\beta \partial^\alpha \mathcal{J}_{\delta}^{\delta\epsilon} + 2 \partial_\epsilon \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}_\chi^{\delta\epsilon} - 3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\beta\delta\epsilon} -$ $3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\alpha\delta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\alpha\beta\delta} + \eta^{\alpha\beta} \partial_\phi \partial_\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\delta\epsilon\phi} - \eta^{\alpha\beta} \partial_\phi \partial^\phi \partial_\epsilon \partial_\chi \mathcal{J}_{\delta}^{\delta\epsilon}) = 0$	
Total # constraint(s):	5		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{\overset{(4)}{\kappa_3}}$
	1	$-\frac{8 \overset{(4)}{\kappa_3} \overset{(2)}{\kappa_2}}{8 (\overset{(4)}{\kappa_1} + \overset{(4)}{\kappa_2}) \overset{(4)}{\kappa_3} - 3 \overset{(4)}{\kappa_4}^2}$	$\frac{8 \overset{(4)}{\kappa_3}^2 + 12 \overset{(4)}{\kappa_3} \overset{(4)}{\kappa_4} + 9 \overset{(4)}{\kappa_4}^2}{\overset{(4)}{\kappa_3} (8 \overset{(4)}{\kappa_1} \overset{(4)}{\kappa_3} + 8 \overset{(4)}{\kappa_2} \overset{(4)}{\kappa_3} - 3 \overset{(4)}{\kappa_4}^2)}$
	3	$\frac{16 \overset{(4)}{\kappa_3} \overset{(2)}{\kappa_2}}{-16 \overset{(4)}{\kappa_2} \overset{(4)}{\kappa_3} + 4 \overset{(4)}{\kappa_3}^2 + 4 \overset{(4)}{\kappa_3} \overset{(4)}{\kappa_4} + 5 \overset{(4)}{\kappa_4}^2}$	$\frac{3 (12 \overset{(4)}{\kappa_3}^2 + 20 \overset{(4)}{\kappa_3} \overset{(4)}{\kappa_4} + 15 \overset{(4)}{\kappa_4}^2)}{2 \overset{(4)}{\kappa_3} (-16 \overset{(4)}{\kappa_2} \overset{(4)}{\kappa_3} + 4 \overset{(4)}{\kappa_3}^2 + 4 \overset{(4)}{\kappa_3} \overset{(4)}{\kappa_4} + 5 \overset{(4)}{\kappa_4}^2)}$
Resolved unitarity condition(s):		(Demonstrably impossible)	